






Dr Govind Narayan Purohit
Theriogenology
GNP SIR CLASSES

Livestock Production and Management -I

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Demographic distribution of livestock and role in Indian economy

Livestock Management

Livestock means stock (item of property)that is live and used for production of items for commerce and /or domestic consumption. The item “**Livestock**”, in the broad sense , includes all animals , birds and other living creatures used for producing products for the use of man. But in the narrow sense, the item “**Livestock**” , is loosely used to include the mammal farm animals.

presently the term “Livestock production ” and “animal production” are used synonymously in the country to indicate farm animal production or animal agricultural, as the western call it. Hence, the term “agricultural animals” is used to indicate livestock that is reared by the farmers for the profit or to satisfy family need.

Definition:

Livestock management involves integrated application of the principles of animal breeding, feeding, housing, organization and disease control in a manner suitable for a particular situation.

Animal production involves

1.Nutrition

2.Fodder production

3.Better breeding

4.Regular reproduction

5.Better disease prevention

But better management includes

1.Economic feeding

2.Identification of better breeding stock

3.Maintenance of their records and implementation of mating plan

4.Monitoring the reproductive efficiency

General principles of animal management

The basic requirements for the welfare of livestock are :-

1. Provision of readily accessible fresh water
2. Nutritionally adequate feed as required
3. Provision of adequate temperature and ventilation
4. Adequate freedom for movement and to stretch their body
5. Sufficient light for satisfactory inspection and also for feeding
6. Rapid diagnosis and treatment of injuries and disease
7. Emergency provision in the event of breakdown of essential mechanical equipment
8. Flooring which neither harms nor cause undue stress to the animal
9. Domestication and rearing of animals for production causes considerable strain on the body of the animals.

It is therefore essential that these animals should be looked after well.

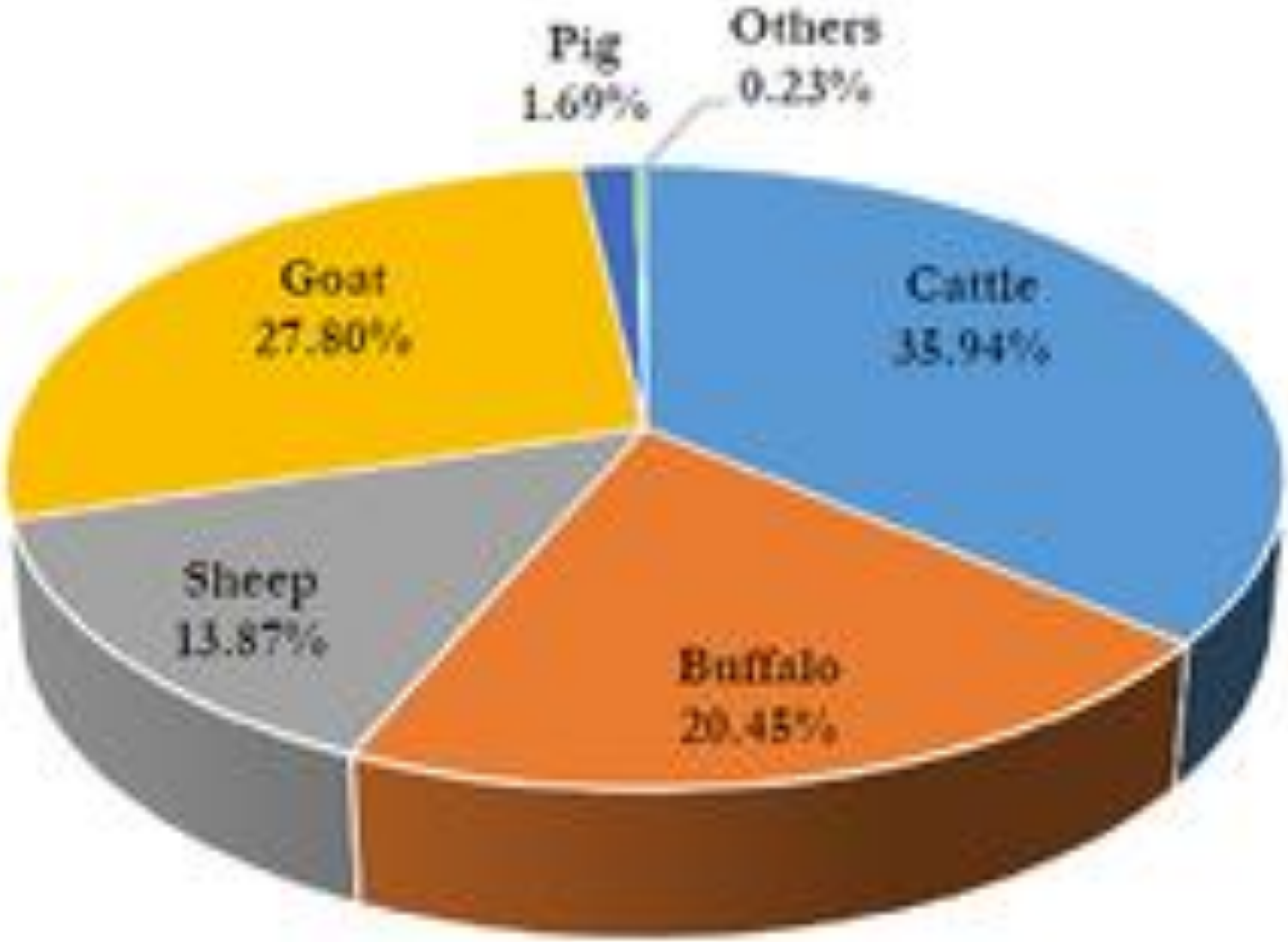
India is:-

1. World’s highest livestock owner at about 535.78 million
2. First in the total buffalo population in the world - 109.85 million buffaloes
3. Second in the population of goats - 148.88 million goats
4. Second largest poultry market in the world
5. Second largest producer of fish and also second largest aquaculture nation in the world
6. Third in the population of sheep (74.26 millions)
7. Fifth in in the population of ducks and chicken (851.81 million)
8. Tenth in camel population in the world - 2.5 lakhs

Category	Population (In million) 2012	Population (In million) 2019	% growth
Cattle	190.90	192.49	0.83
Buffalo	108.70	109.85	1.06
Sheep	65.07	74.26	14.13
Goat	135.17	148.88	10.14
Pig	10.29	9.06	-12.03
Mithun	0.30	0.38	26.66
Yak	0.08	0.06	-25.00
Horses & Ponics	0.63	0.34	-45.58
Mule	0.20	0.08	-57.09
Donkey	0.32	0.12	-61.23
Camel	0.40	0.25	-37.05
Total Livestock	512.06	535.78	4.63

S.No.	States	Population (In million) 2012	Population (In million) 2019	% Change
1	Uttar Pradesh	68.7	67.8	-1.35
2	Rajasthan	87.7	56.8	-1.66
3	Madhya Pradesh	36.3	40.6	11.81
4	West Bengal	30.3	37.4	23.32
5	Bihar	32.9	36.5	10.67
6	Andhra Pradesh	29.4	34.0	15.79
7	Maharashtra	32.5	33.0	1.61
8	Telangana	26.7	32.6	22.21
9	Karnataka	27.7	29.0	4.70
10	Gujarat	27.1	26.9	-0.95

Graph 1: Livestock Population 2019 - Share of Major Species



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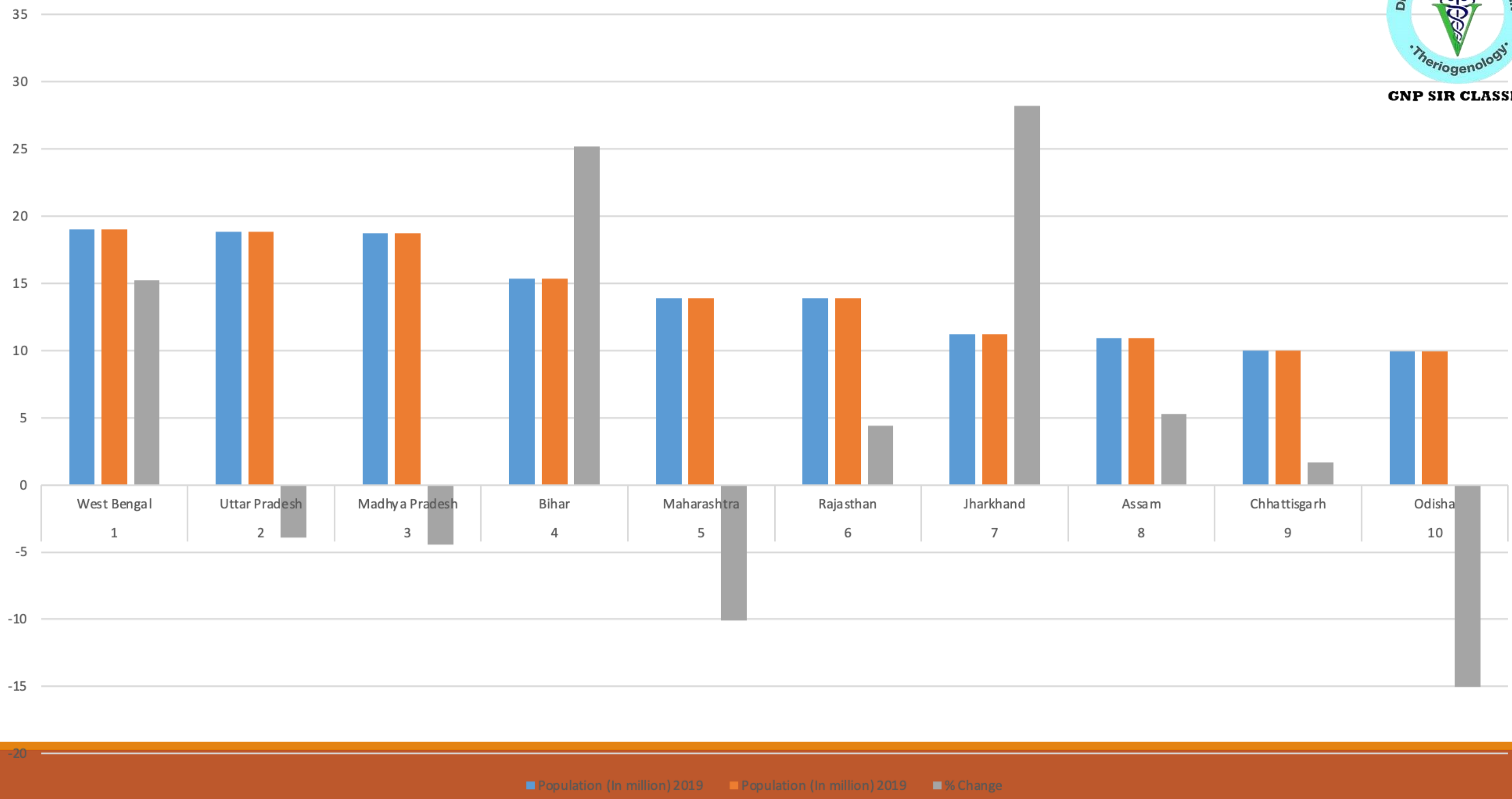
	Population (In million) 2012	Population (In million) 2019	% growth
Total Poultry	729.21	851.81	16.81
Backyard poultry	217.49	317.07	45.78
Commercial Poultry	511.72	534.74	4.50

Poultry Population 2012 and 2019

- The total poultry has increased by 16.81% and the total poultry is 851.81 million during 2019.
- Over 45.78% increase in backyard poultry and total backyard poultry is 317.07 Million in 2019.
- The commercial poultry has increased by 4.5% and the total commercial poultry is 534.74 million.

SL. No.	Product	Quantity	Ranking in the world production
1	MILK in million tonnes	222.30	FIRST
2	EGGS in millions Nos	122,217	THIRD
3	MEAT million tonnes	9.70	NA
4	WOOL in million kgs	41.50	NA
5	FISH in million metric tonnes	12.61	SECOND

Cattle Population, 2012 & 2019 in Major States



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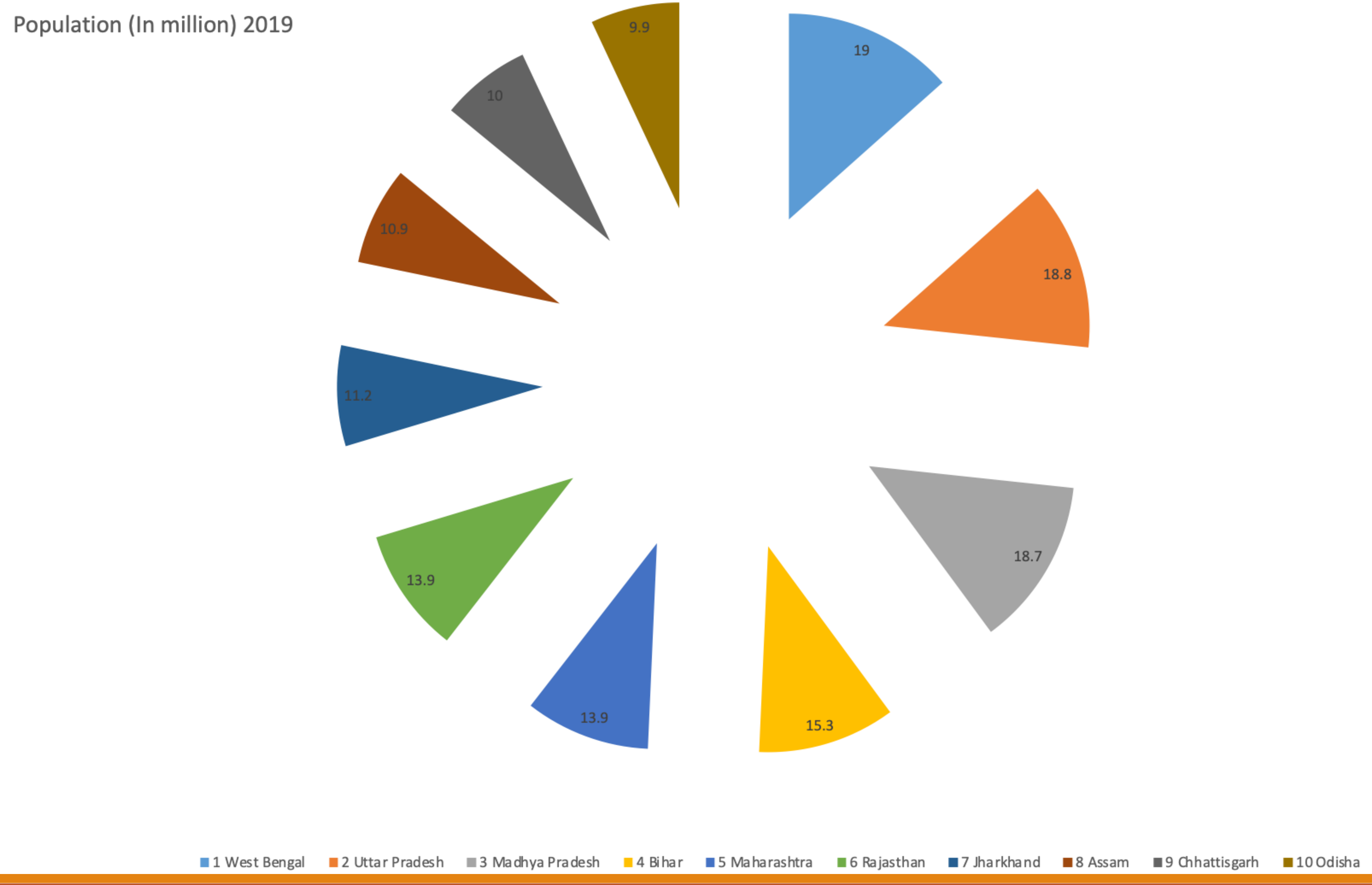
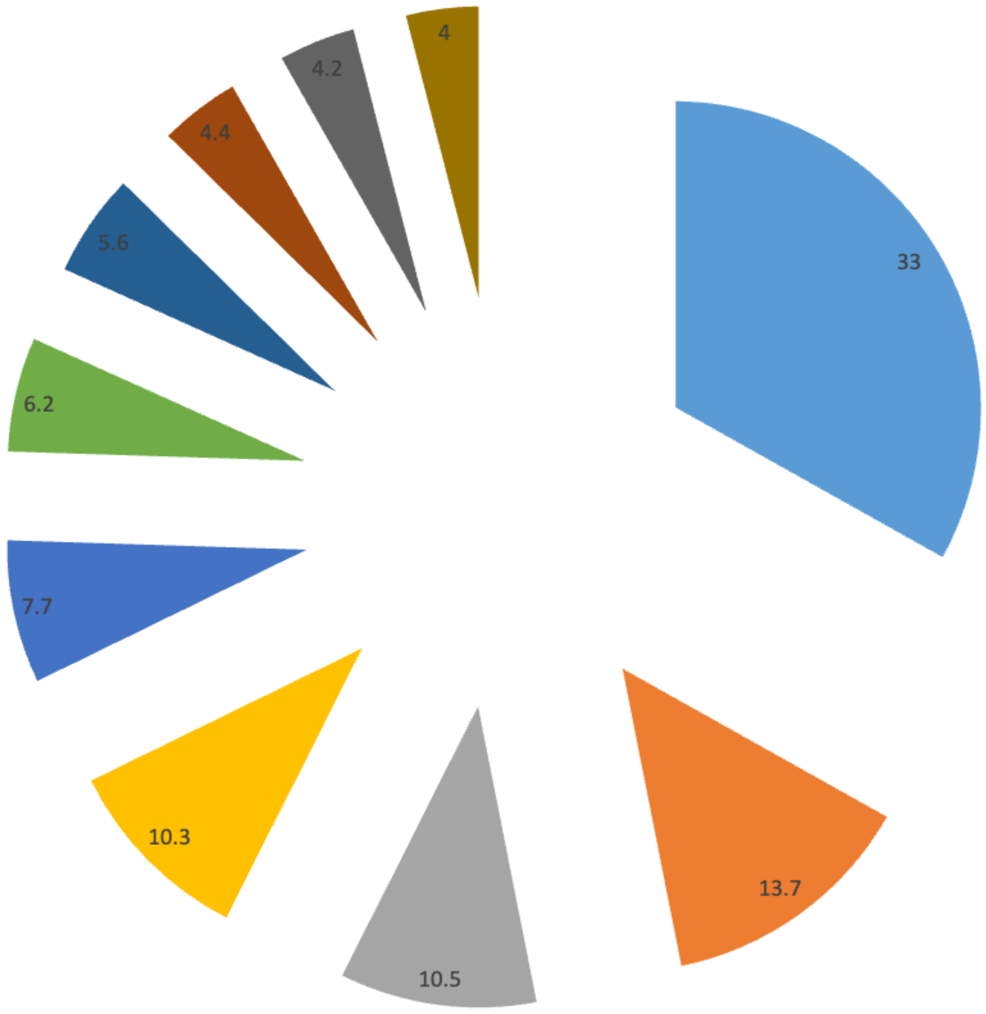

Population (In million) 2019
19
18.8
18.7
15.3
13.9
13.9
11.2
10.9
10
9.9
1 West Bengal
2 Uttar Pradesh
3 Madhya Pradesh
4 Bihar
5 Maharashtra
6 Rajasthan
7 Jharkhand
8 Assam
9 Chhattisgarh
10 Odisha

Buffalo Population 2012 & 2019 of Major States

Population (In million) 2019



1 Uttar Pradesh 2 Rajasthan 3 Gujarat 4 Madhya Pradesh 5 Bihar 6 Andhra Pradesh 7 Maharashtra 8 Haryana 9 Telangana 10 Punjab

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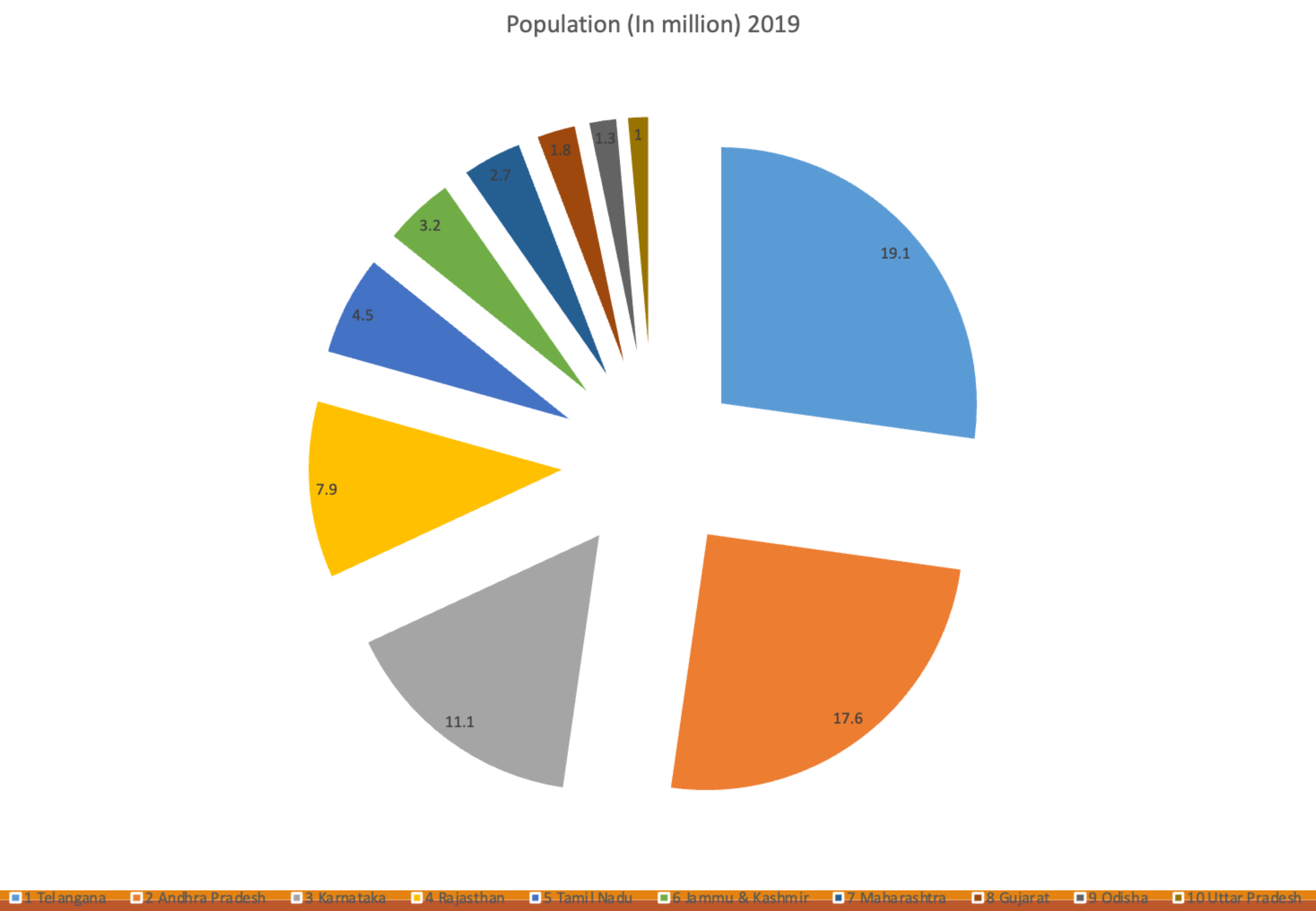
Sheep Population 2012 & 2019 of Major States



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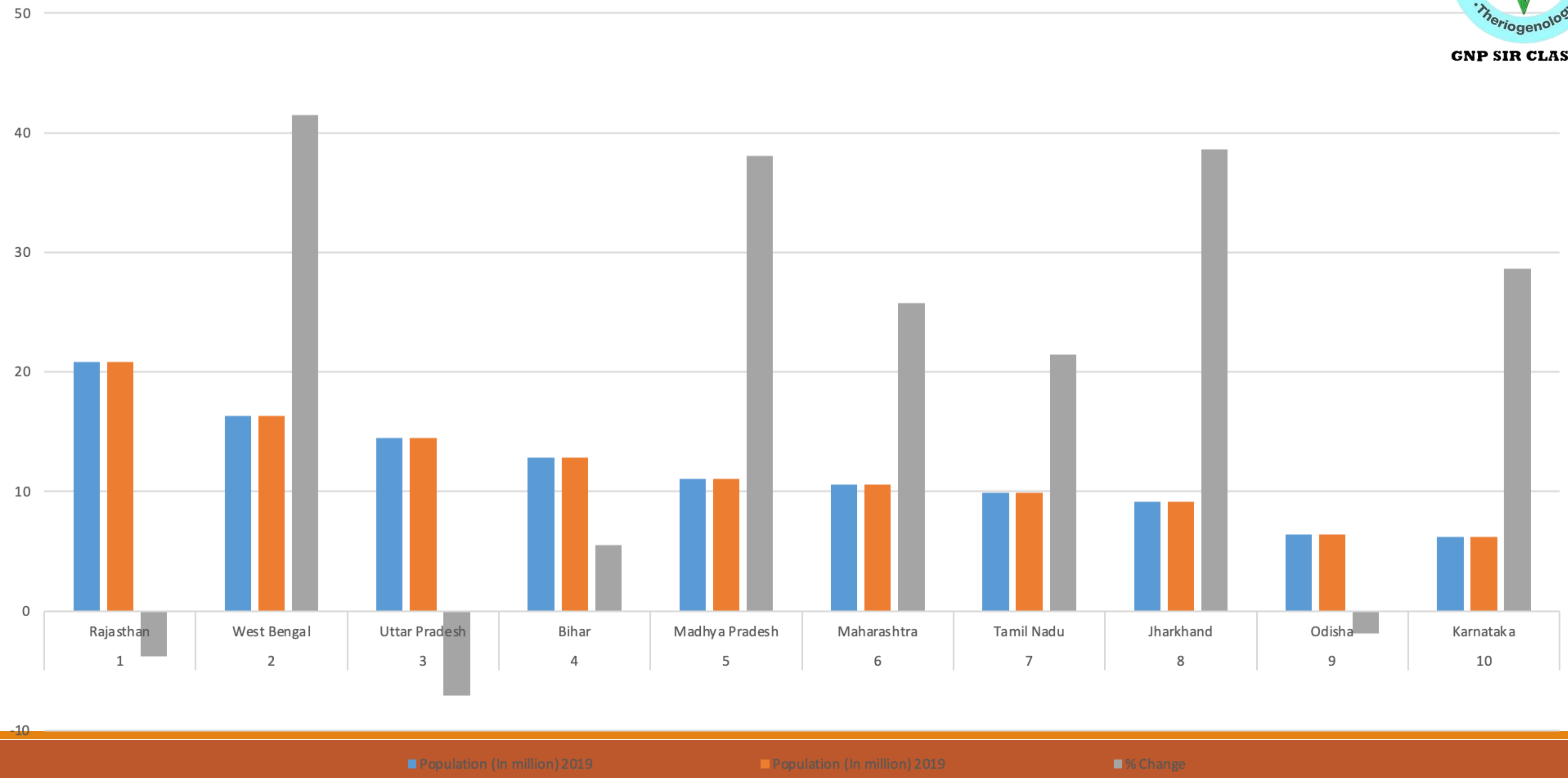
Goat Population 2012 & 2019 of Major States



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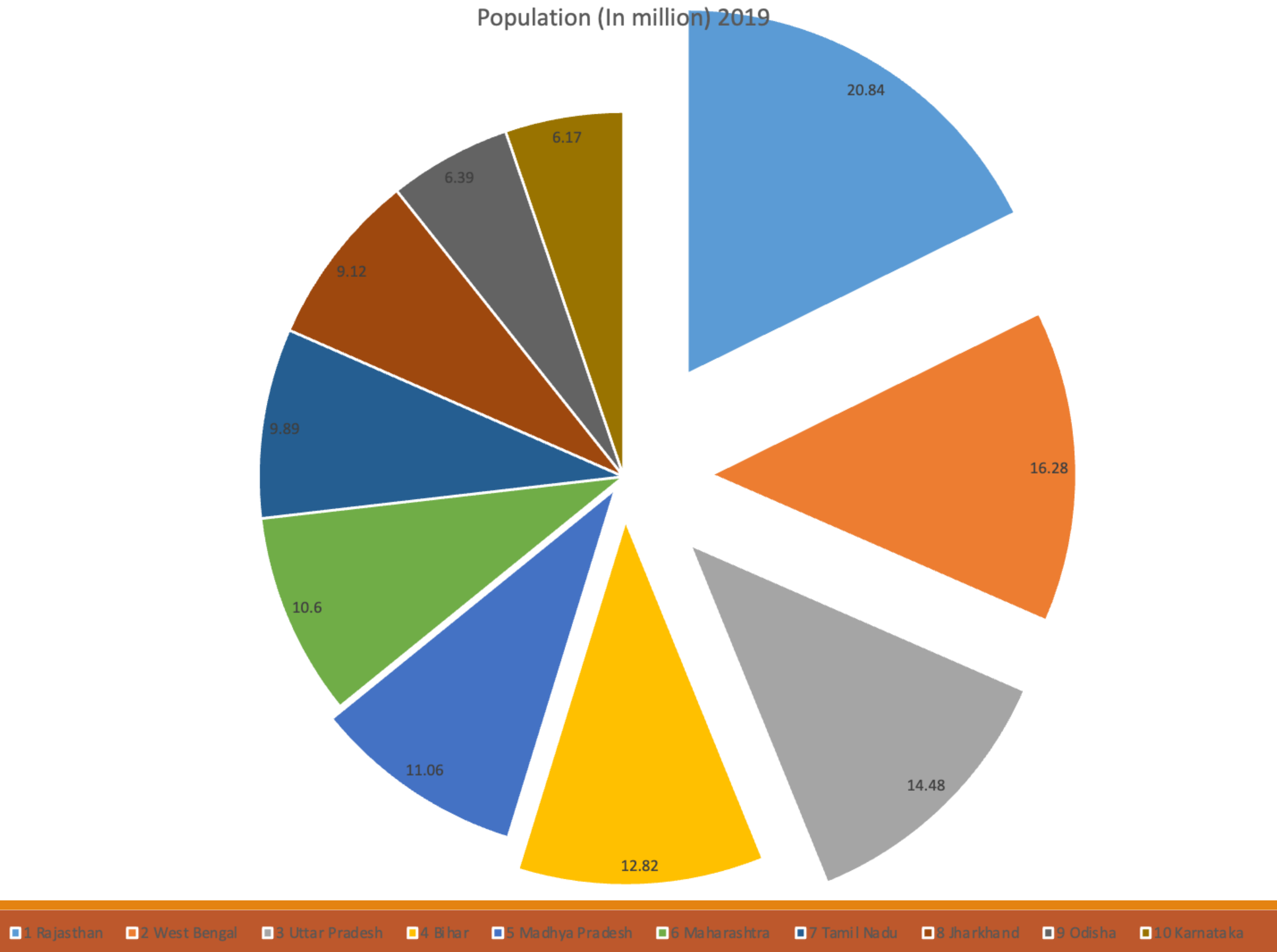

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Pig :

- Total Pig Population in the country is 9.06 Million during 2019.
- Total Pig Population has decreased by 12.0% over previous Livestock Census (2012).
- About 1.7% of the total livestock is contributed by pigs.

Camel:

- Total Camel Population in the country is 0.25 Million during 2019.
- Total Camel Population has decreased by 37.1% over previous Livestock Census (2012).

Horses, Ponies, Mules & Donkeys:

- Total Population of Horses, Ponies, Mules & Donkeys in the country is 0.55 Million during 2019.
- Total Population of Horses, Ponies, Mules & Donkeys has decreased by 51.9% over previous Livestock Census (2012).

Mithun:

- Total Population of Mithun in the country is 3.9 Lakhs during 2019.
- Total Population of Mithun has increased by 30.0% over previous Livestock Census (2012).

Yak:

- Total Population of Yak in the country is 58 Thousands during 2019.
- Total Population of Yak has decreased by 24.67% over previous Livestock Census (2012).

1. The total Livestock population is **535.78** million in the country showing an increase of **4.6%** over Livestock Census 2012
2. Total Bovine population (Cattle, Buffalo, Mithun and Yak) is **302.79 Million** in 2019 which shows an increase of about **1%** over the previous census.
3. The total number of cattle in the country in 2019 is **192.49 million** showing an increase of **0.8 %** over previous Census.
4. The Female Cattle (Cows population) is **145.12 million**, increased by **18.0%** over the previous census (2012).
5. The Exotic/Crossbred and Indigenous/Non-descript Cattle population in the country is **50.42 million** and **142.11 million** respectively.
6. The Indigenous/Non-descript female cattle population has increased by **10%** in 2019 as compared to previous census.
7. The population of the total Exotic/Crossbred Cattle has increased by **26.9 %** in 2019 as compared to previous census.
8. There is a decline of 6 % in the total Indigenous/ Non-descript cattle population over the previous census. However, the pace of decline of Indigenous/ Non-descript cattle population during 2012-2019 is much lesser than as compared to the 2007-12 which was about 9%.
9. The total buffaloes in the country is **109.85 million** showing an increase of about 1.0% over previous Census.
10. The total milch animals (in-milk and dry) in cows and buffaloes is **125.34 million**, an increase of **6.0 %** over the previous census.
11. The total sheep in the country is **74.26 million** in 2019, increased by **14.1%** over previous Census.
12. The Goat population in the country in 2019 is **148.88 million** showing an increase of **10.1%** over the previous census.
13. The total Pigs in the country is **9.06 Million** in the current Census, declined by **12.03%** over the previous Census.
14. The other livestock including mithun, yak, horses, ponies, mule, donkeys, camel together contribute around 0.23% of the total livestock and their total count is **1.24 million**.
15. The total poultry in the country is **851.81 million in 2019**, registered an increase of **16.8%** in the total poultry.
16. The total birds in the backyard poultry in the country is **317.07 million**. The backyard poultry has increased by around **46%** as compared to previous Census.
17. The total Commercial Poultry in the country is **534.74 million** in 2019, increased by **4.5%** over previous Census.

Value of Output from Livestock sector- At Current Prices in Rs. Crore

Item	2011-12	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21
1 Milk Group	3,27,767	3,72,228	4,23,150	4,95,835	5,60,823	6,29,259	7,07,607	7,75,574	8,47,058	9,31,969
2 Meat Group	96,219	1,14,995	1,36,663	1,54,152	1,71,636	2,06,691	2,22,131	2,51,165	2,92,614	3,34,789
2.1 Meat	88,469	1,05,828	1,26,211	1,42,195	1,58,640	1,92,015	2,06,607	2,33,744	2,72,211	3,10,211
2.1.1 Beef	11,754	14,324	16,596	20,587	25,020	26,534	27,716	30,964	33,471	35,054
2.1.2 Mutton	32,667	38,878	44,212	50,176	53,396	63,184	68,362	77,046	94,703	1,23,177
2,1,3 Pork	4,465	4,886	5,783	6,119	5,431	6,637	6,209	6,597	7,476	7,481
2.1.4 Poultry Meat	39,583	47,740	59,620	65,313	74,793	95,660	1,04,320	1,19,137	1,36,560	1,44,499
2.2 Meat Products	3,346	3,975	4,539	5,244	5,649	6,557	7,016	7,883	9,477	11,887
2.3 By-Products	4,404	5,192	5,914	6,713	7,347	8,120	8,508	9,538	10,927	12,691
2.3.1 Hides	1,887	2,213	2,546	2,852	3,194	3,325	3,346	3,783	4,004	3,898
2.3.2 Skins	1,189	1,396	1,561	1,760	1,867	2,153	2,329	2,567	3,098	4,003
2.3.3 Other By Products	1,328	1,583	1,807	2,101	2,287	2,642	2,832	3,188	3,824	4,789
3 Eggs	16,633	19,690	22,708	24,382	26,657	29,777	32,513	36,592	42,224	47,785
4 Wool & hair	496	539	584	578	541	628	606	666	662	630
5 Dung	32,599	40,453	43,079	45,455	48,478	53,754	56,208	58,537	64,035	68,715
6 Silk Worm Cocoonsand Honey	4,326	4,947	6,286	6,930	7,606	8,945	10,175	10,973	11,619	11,613
7 Increment in Stock	9,710	12,086	13,708	15,474	17,757	21,624	13,839	16,276	17,780	19,089
Value of Output from Livestock Sector	4,87,751	5,64,937	6,46,178	7,42,807	8,33,498	9,50,677	10,43,079	11,49,783	12,75,993	14,14,590

Source : National Accounts Statistics 2022, Central Statistical Organisation, GoI

State-wise Estimates of Annual Production of MLPs with Growth Rate during 2020-21* and 2019-20

S.No.	State/UT	Milk ('000 Tns.) 2020-21*	Milk ('000 Tns.) 2019-20	Growth Rate (%)	Egg (Lakh Nos.) 2020-21*	Egg (Lakh Nos.) 2019-20	Growth Rate (%)
1	Andhra Pradesh	14713.84	15263.10	-3.60	249639.17	219275.18	13.85
2	Arunachal Pradesh	43.89	60.63	-27.61	638.51	604.92	5.55
3	Assam	954.07	919.94	3.71	5253.04	5148.75	2.03
4	Bihar	11501.58	10480.33	9.74	30132.16	27408.02	9.94
5	Chhattisgarh	1747.28	1675.56	4.28	19787.84	20289.04	-2.47
6	Goa	60.14	61.43	-2.11	400.18	400.02	0.04
7	Gujarat	15852.69	15292.35	3.66	19319.41	19274.18	0.23
8	Haryana	11283.55	11734.72	-3.84	72733.93	66153.17	9.95
9	Himachal Pradesh	1576.44	1530.94	2.97	1111.00	1066.21	4.20
10	Jammu & Kashmir	2594.50	2506.36	3.52	2128.27	2216.19	-3.97
11	Jharkhand	2434.02	2321.16	4.86	7843.28	6928.05	13.21
12	Karnataka	10936.44	9031.49	21.09	76199.44	66511.22	14.57
13	Kerala	2533.88	2544.35	-0.41	22134.89	21845.10	1.33
14	Madhya Pradesh	17999.30	17108.96	5.20	26515.73	23793.88	11.44
15	Maharashtra	13703.32	12024.26	13.96	64028.47	63713.01	0.50
16	Manipur	71.05	89.65	-20.74	1141.63	1081.81	5.53
17	Meghalaya	89.14	87.60	1.76	1107.85	1102.02	0.53
18	Mizoram	25.38	23.53	7.85	434.06	433.38	0.16
19	Nagaland	55.93	61.64	-9.27	384.87	381.64	0.85
20	Orissa	2372.81	2370.09	0.11	24242.67	23814.02	1.80
21	Punjab	13393.99	13347.56	0.35	56511.64	56387.99	0.22
22	Rajasthan	30723.11	25573.09	20.14	27982.86	26962.40	3.78
23	Sikkim	73.85	83.94	-12.01	98.13	48.27	103.28
24	Tamil Nadu	9790.36	8759.01	11.77	201207.87	200216.03	0.50
25	Telangana	5765.21	5589.80	3.14	158470.11	148055.17	7.03
26	Tripura	206.17	198.60	3.81	3031.68	2949.81	2.78
27	Uttar Pradesh	31359.10	31863.91	-1.58	36288.91	34049.29	6.58
28	Uttarakhand	1797.45	1844.71	-2.56	4924.14	4786.30	2.88
29	West Bengal	6164.82	5868.61	5.05	105008.00	97350.10	7.87
30	A&N Islands	14.67	18.90	-22.40	1361.84	1188.86	14.55
31	Chandigarh	52.30	48.87	7.02	130.08	124.50	4.48
32	Ladakh	14.88	0.00	-	12.76	0.00	-
33	D.& N. Haveli	1.40	-	15.47	4.99	-	-4.15
	Daman & Diu		1.21			5.21	
34	Delhi	-	-	-	-	-	-
35	Lakshadweep	3.81	3.78	0.91	163.23	153.48	6.36
36	Puducherry	49.50	49.50	0.00	112.70	112.70	0.00

Share of Agriculture & Allied and Livestock Sector in GVA (₹ Crore)					
At Current Prices					
Year	GVA (Total)	GVA (Agriculture & Allied)		GVA (Livestock Sector)	
		Amount	% Share to total GVA	Amount	% Share to total GVA
2011-12	81,06,946	15,01,947	18.5	3,27,334	4.0
2012-13	92,02,692	16,75,107	18.2	3,68,823	4.0
2013-14	1,03,63,153	19,26,372	18.6	4,22,733	4.1
2014-15	1,15,04,279	20,93,612	18.2	5,10,411	4.4
2015-16	1,25,74,499	22,27,533	17.7	5,82,410	4.6
2016-17	1,39,65,200	25,18,662	18.0	6,72,611	4.8
2017-18	1,55,05,665	28,29,826	18.3	7,85,683	5.1
2018-19	171,75,128	30,29,925	17.6	8,82,009	5.3
2019-20	183,55,109	33,58,364	18.3	9,77,730	5.2
2020-21	180,57,810	36,09,494	20.0	11,14,249	6.2

Source : National Accounts Statistics-2022, Central Statistical Organisation, GoI

State-wise Per Capita Availability of Milk during 2019-20 and 2020-21* (in Gram/Day)

Sl.No.	State/UT	2019-20	2020-21*	Change
1	Andhra Pradesh	799	768	-31
2	Arunachal Pradesh	110	79	-31
3	Assam	73	75	2
4	Bihar	240	260	20
5	Chhattisgarh	159	164	5
6	Goa	109	106	-3
7	Gujarat	615	631	16
8	Haryana	1118	1063	-55
9	Himachal Pradesh	573	588	15
10	Jammu & Kashmir+ Ladakh	507	534	27
11	Jharkhand	170	176	6
12	Karnataka	375	452	77
13	Kerala	198	197	-1
14	Madhya Pradesh	568	591	23
15	Maharashtra	269	305	36
16	Manipur	79	62	-17
17	Meghalaya	74	75	1
18	Mizoram	54	58	4
19	Nagaland	78	71	-7
20	Orissa	144	143	-1
21	Punjab	1221	1219	-2
22	Rajasthan	904	1075	171
23	Sikkim	345	302	-43
24	Tamil Nadu	316	353	37
25	Telangana	410	422	12
26	Tripura	136	140	4
27	Uttar Pradesh	387	377	-10
28	Uttarakhand	447	437	-10
29	West Bengal	165	173	8
30	A&N Islands	130	101	-29
31	Chandigarh	113	120	7
32	Ladakh	-	138	-
33	Dadra & Nagar Haveli	-	4	-4
	Daman & Diu	8		
34	Delhi	-	-	-
35	Lakshadweep	152	154	2

State-wise Per Capita Availability of Egg during 2019-20 and 2020-21* (in Nos./Annum)

Sl.No.	State/UT	2019-20	2020-21*	Change
1	Andhra Pradesh	420	475	55
2	Arunachal Pradesh	40	42	2
3	Assam	15	15	0
4	Bihar	23	25	2
5	Chhattisgarh	71	68	-3
6	Goa	26	26	0
7	Gujarat	28	28	0
8	Haryana	231	250	19
9	Himachal Pradesh	15	15	0
10	Jammu & Kashmir+ Ladakh	16	16	0
11	Jharkhand	19	21	2
12	Karnataka	101	115	14
13	Kerala	62	63	1
14	Madhya Pradesh	29	32	3
15	Maharashtra	52	52	0
16	Manipur	35	36	1
17	Meghalaya	34	34	0
18	Mizoram	36	36	0
19	Nagaland	18	18	0
20	Orissa	53	53	0
21	Punjab	189	188	-1
22	Rajasthan	35	36	1
23	Sikkim	7	15	8
24	Tamil Nadu	265	265	0
25	Telangana	398	423	25
26	Tripura	74	75	1
27	Uttar Pradesh	15	16	1
28	Uttarakhand	42	44	2
29	West Bengal	100	108	8
30	A&N Islands	299	341	42
31	Chandigarh	11	11	0
32	Ladakh	-	4	-
33	Dadra & Nagar Haveli	0	0.5	-0.5

State-wise Per Capita Availability of Meat during 2019-20 and 2020-21* (in Kg./Annum)

Sl.No.	State/UT	2019-20	2020-21	Change
1	Andhra Pradesh	16.28	18.17	1.89
2	Arunachal Pradesh	15.08	14.7	-0.38
3	Assam	1.54	1.57	0.03
4	Bihar	3.21	3.27	0.06
5	Chhattisgarh	2.3	1.61	-0.69
6	Goa	4.68	3.91	-0.77
7	Gujarat	0.49	0.48	-0.01
8	Haryana	19.32	20.52	1.2
9	Himachal Pradesh	0.65	0.59	-0.06
10	Jammu & Kashmir+ Ladakh	7.07	6.63	-0.44
11	Jharkhand	1.8	1.85	0.05
12	Karnataka	4.63	5.49	0.86
13	Kerala	12.89	13.07	0.18
14	Madhya Pradesh	1.3	1.4	0.1
15	Maharashtra	9.33	8.99	-0.34
16	Manipur	9.25	9.35	0.1
17	Meghalaya	14.38	12.09	-2.29
18	Mizoram	13.77	12.51	-1.26
19	Nagaland	14.93	11	-3.93
20	Orissa	4.55	4.7	0.15
21	Punjab	8.32	7.43	-0.89
22	Rajasthan	2.58	2.72	0.14
23	Sikkim	5.67	4.66	-1.01
24	Tamil Nadu	8.76	8.8	0.04
25	Telangana	22.79	24.56	1.77
26	Tripura	12.67	12.94	0.27
27	Uttar Pradesh	5.18	4.55	-0.63
28	Uttarakhand	2.24	1.65	-0.59
29	West Bengal	9.32	10.19	0.87
30	A&N Islands	14.17	14.15	-0.02
31	Chandigarh	0.68	0.55	-0.13
32	Ladakh		0.06	0.06
33	Dadra & Nagar Haveli	-	0.3	-0.22
	Daman & Diu	0.52		
34	Delhi	-	-	-

Categorisation of Farmers

In agriculture Census, the operational holdings are categorised in five size classes as follows:-

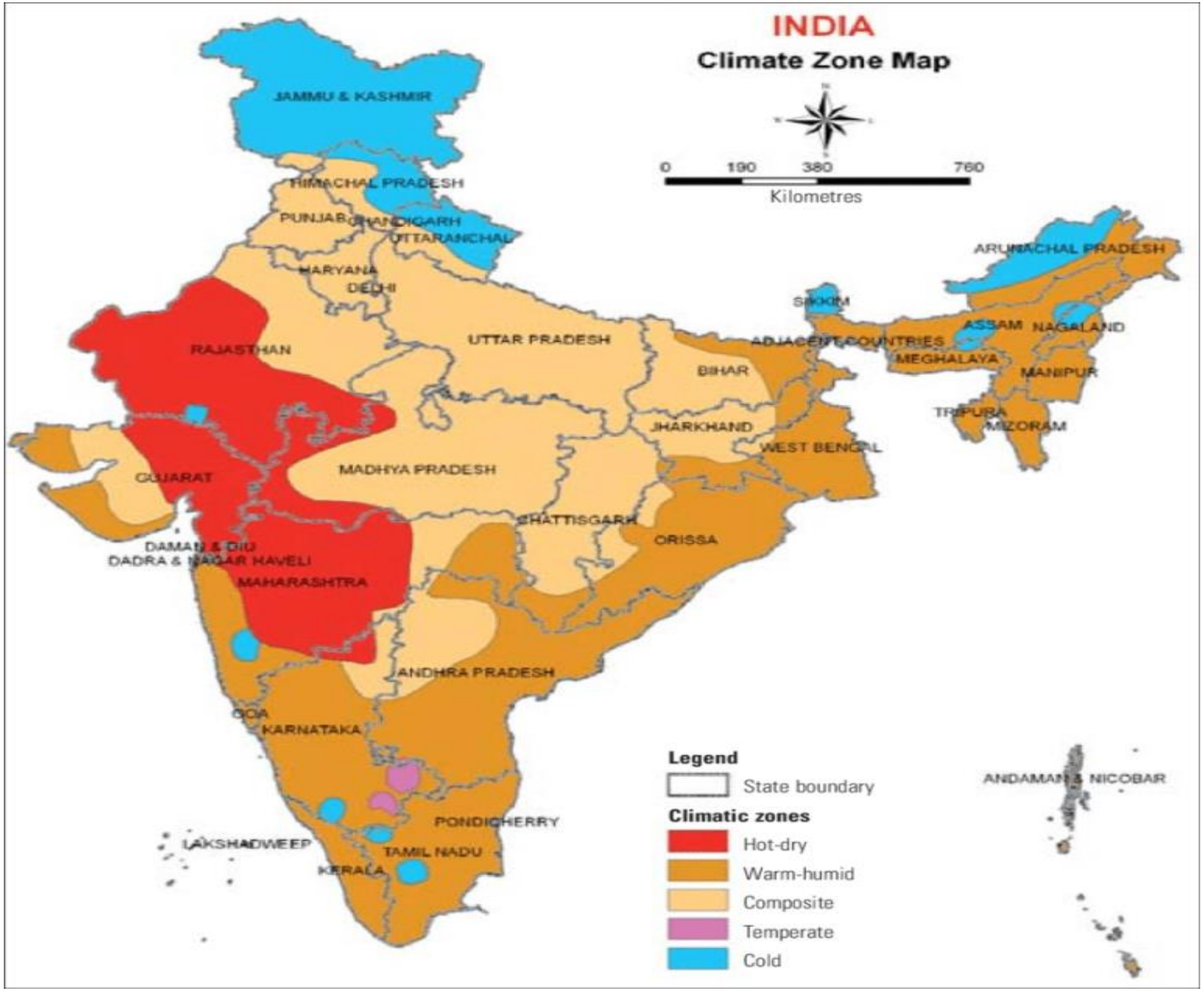
SL. NO.	CATEORY	SIZE-CLASS
1.	Marginal	Below 1.00 hectare
2.	Small	1.00-2.00 hectare
3.	Semi- Medium	2.00-4.00 hectare
4.	Medium	4.00-10.00 hectare
5.	Large	10.00 hectare and above

- **Livestock distribution pattern as per land holdings**
- Total area of operational land holding: 159.44 million ha.
- Total number of operational land holdings: 119.93 million
- Average size of land holding: 1.23 ha.

Category of land holding	% distribution of livestock	Average area per holding (ha)	Cattle & Buffalo/ holding	Sheep & Goat/ holding
Marginal (< 1 ha.)	50.1	0.38	1.45	0.82
Small(1-1.99 ha)	21.6	1.38	2.30	1.43
Semi-medium (2-3.99 ha)	15.9	2.68	2.99	1.89
Medium (4-9.99 ha)	9.6	5.74	4.07	2.67
Large (>10 ha)	2.9	17.08	5.33	4.55
Overall		1.23	1.94	1.17

Land and Animal Holding Patterns in India

CATEGORY	FARMERS (%)	PERCENTAGE OF		MILK PRODUCTION (%)
		Land owned	Milch animals	
Landless agricultural workers	26.0	-	22.5	22.6
Small and marginal farmers	22.5	27.0	41.8	41.9
Medium and large farmers	22.6	73.0	35.7	35.5



GNP SIR CLASSES

1. Hot and Dry Climate

A hot and Dry climate is seen in western parts of the country where desert-like conditions exist.

- Jaipur
- Jaisalmer
- Kutch
- Gujarat
- Maharashtra

Characteristics:

- **Temperatures**: Very hot weather in summer and cold in winter
    - Summer- temperatures vary from 40 to 45 deg C
    - Winters – Min temperatures- 5-25 deg C
    - High-temperature difference between Day and Night
    - High Solar Radiation causes glare
    - Hot Winds
- **Rainfall**: Minimal rainfall
- **Humidity:** Low Relative Humidity
- **Groundcover:** Dry sandy or rocky ground with less vegetation
- **Sky Condition:** Cloudless sky

2. Warm and Humid Climate

Warm and Humid climate is present in regions that are in proximity to the water bodies, such as coastal areas.

- Kerala
- Tamil Nadu
- Goa
- Parts of Andhra Pradesh
- Parts of Orissa

Characteristics:

- **Temperatures:** Not very high
    - Vary between 25-35 deg C in summer and 20-30 deg C in winter.
- **Humidity**: Very high humidity
- **Rainfall:** High rainfall (about 12mm per year)
- **Sky Condition**: Clouds cover 40-80%

3. Composite Climate

The Composite Climate covers the central part of India.

- •New Delhi
- •Kanpur
- •Allahabad

Characteristics:

- •**Temperatures:** 32-43 deg C during summer and 10-25 deg C during winter.
- •**Humidity**: Variable (20-25% in dry periods and 50-95% in wet periods.
- •**Rainfall**: Variable (500-1300mm per year)
- •**Sky Condition:** variable

4. Temperate/Moderate Climate

Temperate/Moderate Climate regions vary in small parts of India and are considered the most comfortable climate zones in India.

- •Banglore
- •Pune

Characteristics

- •**Temperatures:** The temperatures are in comfort ranges between 30-34deg C during Summer and 27-33deg C during winter.
- •**Humidity:** Low in winter and summer (20-55%) and high (55-90%) during monsoons.
- •**Rainfall:** low rainfall (1000mm per year)
- •**Sky Condition:** Mainly clear or dense low clouds in summer.

5. Cold Climate

Cold Climate exists in the northern parts of the country near the Himalayas.

•Shimla

•Shillong

•Leh

Characteristics:

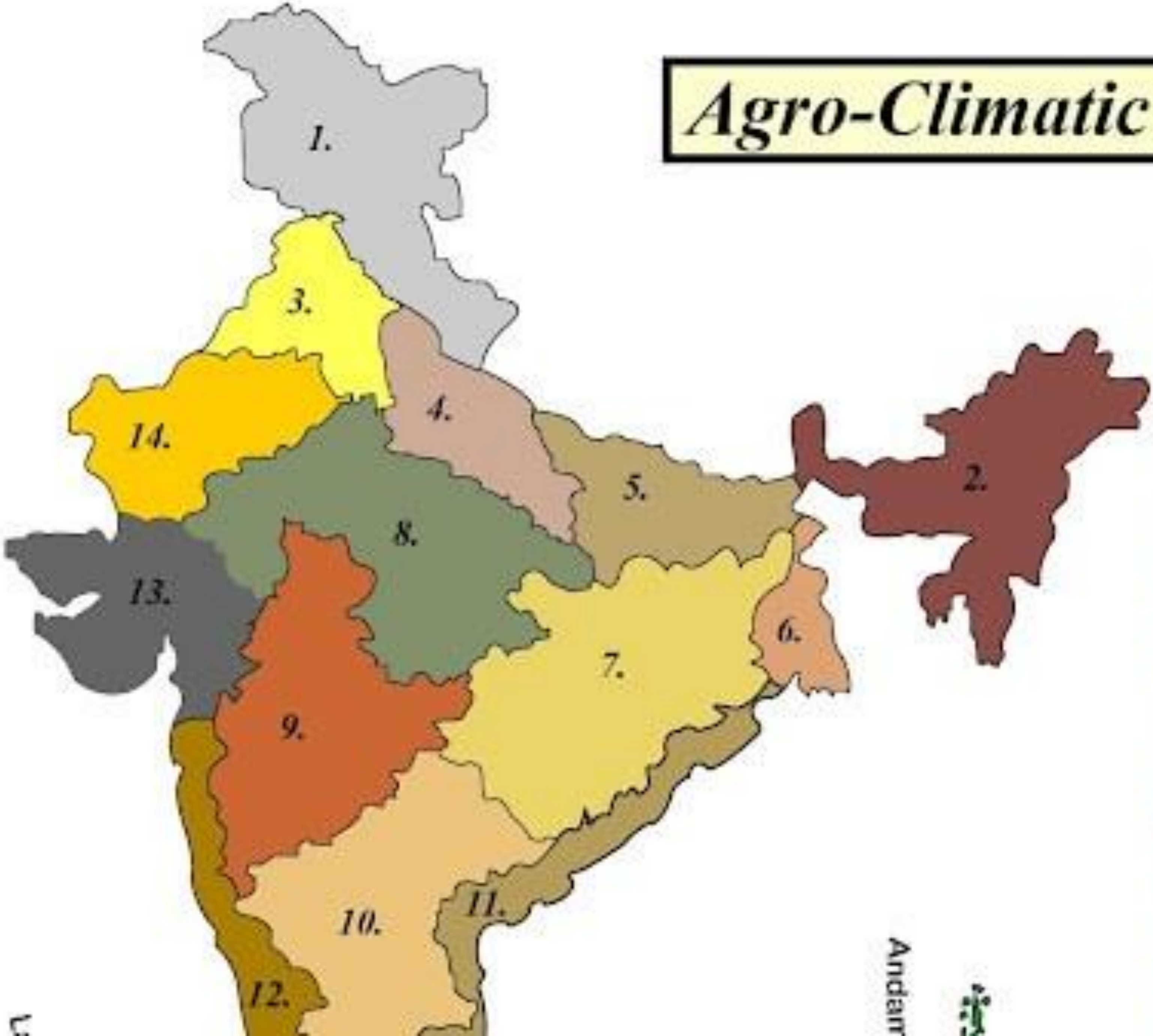
•**Temperatures:** Typically, low temperatures; are 17-20 ^{0}C during summer and -7 to 8 ^{0}C during winter.

•**Humidity:** low in sunny regions and high in cloudy regions.

•**Rainfall:** low (200mm per year) to Moderate (1000mm per year.

•**Sky Condition:** clean with less than 50% cloud cover.

Agro-Climatic Regions in India



1. North-Western Mountain Region
2. The North-Eastern Hill Region
3. The Satluj-Yamuna Plain Region
4. The Upper-Ganga Plain Region
5. The Middle-Ganga Plain Region
6. The Lower-Ganga Plain Region
7. The South-Eastern Plateau Region
8. The Aravali-Malwa Upland
9. The Maharashtra Plateau Region
10. The Deccan Interior Region
11. The Eastern Coastal Region
12. The Western Coastal Region

LIVESTOCK FARMING ZONES OF INDIA

- As per ICAR: 15 agro-climatic regions.

S.No.	Agro-climatic regions/zones	States represented
I	Western Himalayan region	Himachal Pradesh, Jammu & Kashmir, Uttarakhand
II	Eastern Himalayan region	Arunachal Pradesh, Assam, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim, Tripura, West Bengal
III	Lower Gangetic plain region	West Bengal
IV	Middle Gangetic plain region	Uttar Pradesh, Bihar
V	Upper Gangetic plain region	Uttar Pradesh
VI	Trans Gangetic plain region	Chandigarh, Delhi, Haryana, Punjab, Rajasthan
VII	Eastern plateau and hills region	Chhattisgarh, Jharkhand, Madhya Pradesh, Maharashtra, Orissa, West Bengal
VIII	Central plateau and hills region	Madhya Pradesh, Rajasthan, Uttar Pradesh
IX	Western plateau and hills region	Madhya Pradesh, Maharashtra
X	Southern plateau and hills region	Andhra Pradesh, Karnataka, Tamil Nadu

PREDOMINANT LIVESTOCK SPECIES IN DIFFERENT REGIONS

Species	ACZ
Cattle	EPH (7), LGP (3), MGP (4), WCPH (12)
Buffalo	TGP (6), UGP (5), ECPH (11)
Sheep	WD (14), CPH (8), SPH (10), WH (1)
Goats	LGP (3)
Pigs (Moderately)	EH (2), TGP (6), UGP (5), MGP (4), LGP (3),

SI.No.	REGION1	SPECIES PRIORITY				REGIONS SPECIE S PROFIL E	REGION'S PRODUCTY PRIORITY
		Very high	High	Low	Very low		
1	Western Himala yan	-	Shee p	cattle, buffalo, goat	pig	sheep- goat- cattle	Wool-meat
2	Eastern Himalayan	-	cattle , pig	goat	buffalo, sheep	cattle- goat-pig	Meat-draft
3	Lower Gangetic plains	cattle, goat, pig	Sheep	buffalo	-	cattle- goat-pig	Draft-meat
4	Middle Gangetic plains	pig	cattle , buffalo, goat	sheep	-	pig- bovine- goat	Draft-meat
5	Upper Gangetic plains	buffalo, pig	Goat	sheep	-	buffalo- pig-goat	Milk-meat
6	Trans Gangetic plains	buffalo	sheep , pig	cattle, goat	-	buffalo- sheep- goat	Milk-meat
7	Eastern plains and hills		Cattle	buffalo, sheep/g oat, pig	-	cattle-all other	Draft-meat
8	Central plateau and hills	sheep , goat	buffalo	cattle, pig	-	sheep- goat- buffalo	Wool-meat
9	Western plateau and hills	-	-	cattle, buffalo, sheep, goat	-	all livestock	Supplemen tary
10	Southern plateau and hills	sheep	-	cattle, buffalo, sheep, goat	-	sheep-all livestock	Meat- supplemen tary
11	East coast plains and hills	-	buffal o, sheep	cattle, goat, pig	-	buffalo- sheep	Milk- supplemen tary
12	West coast plains and hills	cattle	Goat	pig	buffalo, sheep	cattle- goat-pigs	Draft-meat
13	Gujarat plains and hills	-	-	cattle, buffalo, sheep, goat	pig	all livestock	Supplemen tary
14	Western dry	sheep	-	buffalo, goat	cattle, pig	Sheep/ot hers	Wool-meat
15	Islands	-	-	goat, pig	cattle, buffalo/sh	low livestock	Supplementary

Livestock production systems of different agro-climatic zones

Livestock production systems are considered to be a subset of farming systems. The classification criteria were limited to three: integration with crops, relation to land and agro-ecological zone.

Solely livestock production systems (L)

Livestock systems in which more than 90 % of dry matter fed to animals comes from rangelands, pastures, annual forages and purchased feeds and less than 10 % of the total value of production comes from non-livestock farming activities.

a) Landless livestock production systems (LL) • Subset of the solely livestock production systems in which less than 10 % of the dry matter fed to animals is farm-produced and in which annual average stocking rates are above ten livestock units (LU) per hectare of agricultural land.

b) Grassland-based systems (LG) • Subset of solely livestock production systems in which more than 10 % of the dry matter fed to animals is farm-produced and in which annual average stocking rates are less than ten LU per hectare of agricultural land.

Mixed-farming systems (M) • Livestock systems in which more than 10 % of the dry matter fed to animals comes from crop by-products or stubble or more than 10 % of the total value of production comes from non-livestock farming activities.

1) Rain-fed mixed-farming systems (MR) • A subset of the mixed systems in which more than 90 % of the value of non-livestock farm production comes from rainfed land use.

2) Irrigated mixed-farming systems (MI) • A subset of the mixed systems in which more than 10 % of the value of non-livestock farm production comes from irrigated land use.

Specialized farming:

A specialized farm is one on which 50% or more of the receipts are obtained from a single source. A specialized dairy farm, for example, is one on which the main source of income is milk, while there can also be income from other subsidiary sources like sale of manure, sale of grains and extra fodder etc.

Diversified farming:

A diversified or general purpose farm is one on which the farmer derives income from several commodities or sources. But income from no single source is more than 50% . For example, a farmer may keep a couple of buffaloes (milk) and some chicken (eggs), as well as grow some vegetables and cash crops.

Integrated farming system:

It means the byproducts or waste materials of one enterprise will be used as input materials in another enterprise with the objective of increasing the profit. It is possible even for a resource poor farmer and for him integrated system is more important to maximise the profit through proper utilization of scarce resources in a framing system approach. • In general, husbandry systems are usually classified as intensive, semi-intensive and extensive, but in the tropics and subtropics these distinctions are sometimes less instructive than those between sedentary, transhumant and nomadic systems.

Intensive System:

In intensive system all the operations are confined in one place (shed) and animal movement is restricted. Which facilitate mechanization and more production control. Poultry, pig, rabbit are more suitable for this system. In developed countries dairy also maintained by intensive system.

Semi-intensive system:

In this system the animals are confined during part of a day/night under roof and allowed to graze during day time. During confinement, concentrate feeding is done. This system of rearing is more suitable for dairy, goat and some extent sheep.

Extensive system: In this system, the livestock are kept on grassland; all the operations are in open place. Small shelters are made for young animals during extreme weather conditions.

- In the migratory system sheep and goat farmers make use of the seasonal pastures located in different areas.
- This system is the most common system and applies to all types of ruminants in the Asian region.
- It is characterized by small ruminants, usually owned by small farmers.
- Rearing ruminants is secondary to crop production, consistent with the pattern of agriculture. Buffaloes and cattle tend to be grazed separately, but where goats and sheep are reared, these small ruminants are grazed together, probably because goats tend to lead the herd. The involvement of women and children in rearing small ruminants is very common throughout the Asian regions.

Tethering

as a husbandry system of major importance in the humid zones of this area; it may also be regarded as a semi intensive system practiced by sedentary small farmers, or even as a method of control alternative to fencing or housing.

GNP SIR CLASSES

Sheep production systems

Migrating systems: In India a bulk of sheep population possessed by nomads and tribal in northern temperate region are managed by migrating pattern.

Semi – Migrating systems: Short distance movement within state/ division or district with the change of season in search of grazing grounds is quite common. These sheep farmers invariably have small or marginal land holding. The flocks are taken away during cultivation in the area. The flocks return to their villages after harvest. Harvested field serve a grazing boon and manuring of field for next crop is also possible during the stop over period by night holding of flocks in agricultural land.

Sedentary or Stationary systems: Rearing of small number of sheep on stationary footing near the farmers homestead round the year is called sedentary system. The flock is taken in the morning for grazing and return back to the homestead in the evening. An economic unit size in stationary condition is 30 to 50 sheep.

Goat Production Systems in the Tropics

Extensive-nomadic systems

Extensive - Nomadic systems are the most difficult to improve, because they involve continuous movement, not only of the whole flock, but also of all its owners.

- There is, therefore, no possibility of dividing the flock by age, sex or stage of reproductive cycle, and it is very difficult to provide supplementary feeding.
- Nomadic flockers know where to find the best pasture and browse, as well as drinking water, at different seasons.
- These systems have developed primarily in areas of sparse or seasonal grazing, such as arid regions.

Typical transhumance system

- Typical transhumant system involves the flock spending part of each year grazing within reach of a permanent village base, and the rest of the year on distant pastures, usually in a different ecological zone.
- The women, children and old people usually stay in the village throughout the year, and may cultivate some crops.
- Pregnant and newly kidded does could be kept at the village and fed on crop wastes, tree leaves, etc. being controlled either by tethering or housing.
- Bucks could be similarly controlled, making selective breeding and control of kidding dates possible.
- The transhumance system is practiced in order to locate the best herbage resources from pastures and grasslands.
- There are also well recognized pastoral tribes who practice a complete transhumance, moving from one place to another on traditional migratory routes.
- The transhumant system is prevalent in the Himalayan region.

Sedentary Systems

- Nomadic and transhumance systems are essentially extensive for at least part of the year, but sedentary farmers have a wide choice of systems, from fully extensive to zero-grazing.
- Extensive systems are most appropriate where large areas of pasture land can provide grazing and browse for goats with a minimum of labor or capital investment.
- A more sophisticated method is the running wire in which the tether is attached by a sliding metal ring to a long wire tightly stretched between two short posts.

Semi- intensive system

- Tethering of goats. Goats are usually tethered singly. Where tethering is used, care must be taken that there is no possibility of strangulation by entanglement with vegetation, etc., or with other goats. Shade must be always available, and drinking water and shelter from rain must be provided when required.
- It is essential to change the place of tethering every day for obtaining fresh herbage and a variety of different feed plants by the animal. Tethering is an excellent and cheap method.

Other system of Small ruminant production systems

Systems combining arable cropping:

• Ruminant production systems combining arable cropping have in situations where crop production is important to the contribution to the stability of system. Animals do not compete for the same land, and play a supplementary role to arable cropping

- Road side, Communal and arable grazing systems : Grazing on road side and communal (waste) lands may be practiced by landless stock owners.
- Cut and Carry feeding: In the cut and carry system, a large portion of the feed is usually brought from outside the holding area because of the small size of holdings in relation to the number of animals kept.

Systems integrated with tree cropping

Systems integrated with tree cropping especially common in the humid and sub humid regions where there is intensive crop production. Although the system is not new, integration with tree crops enables more complete utilization of the land.

GNP SIR CLASSES

Problems and prospects of livestock industry in India

Challenges to the livestock industry in India

1. The land is only common resource acting as the platform for the development of human and animals. Ever-increasing competition between human and animal for food is a critical challenge in animal rearing.
2. The improper and unscientific management of resource like water in animal husbandry makes the increased usage of virtual water in the export market. Virtual water is the water embodied in the production of food and fiber and non-food commodities, including energy.
3. The percolation of extension services regarding animal husbandry is less and negligent.
4. The low production potential of the indigenous verities (breeds) of animals is also an added issue.
5. Devastating pastures, increased climatic changes, decrease in monsoon rains also makes the animal husbandry vulnerable
6. Improper awareness about the vaccination and lack of research in the diseases occurring to animals is also another factor.
7. Lack of infrastructure facilities in rural areas like veterinary clinics, breeding centres.
8. The absence of credit facilities and insurance coverage schemes makes the farmers show less interest in animal rearing.
9. The proper quality checking and standardization of animal products is absent.

To be done for livestock industry

1. Culling of burdening animals.
2. Licensing for animal rearing.
3. Rejuvenation of Natural Pastures and Grasslands.
4. Ensure proper credit facilities.
5. Ensure effective veterinary extension services.
6. Promote Research in organic livestock Farming.
7. Special economic zone for the animal husbandry with common facilities must be created for greater export potential.